PREDICTIVE ANALYSIS OF THE PIMA INDIANS DIABETES DATASET

PROJECT FOR THE COURSE: *FOUNDATIONS AND CONCEPTS OF DATA SCIENCE*

NAME: NIKOO NAJAFIAN

SUPERVISORS: DR. SAJJAD ASHRAFI/DR. SARINA MALEKI

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BUSINESS UNDERSTANDING

- **Problem Overview:**

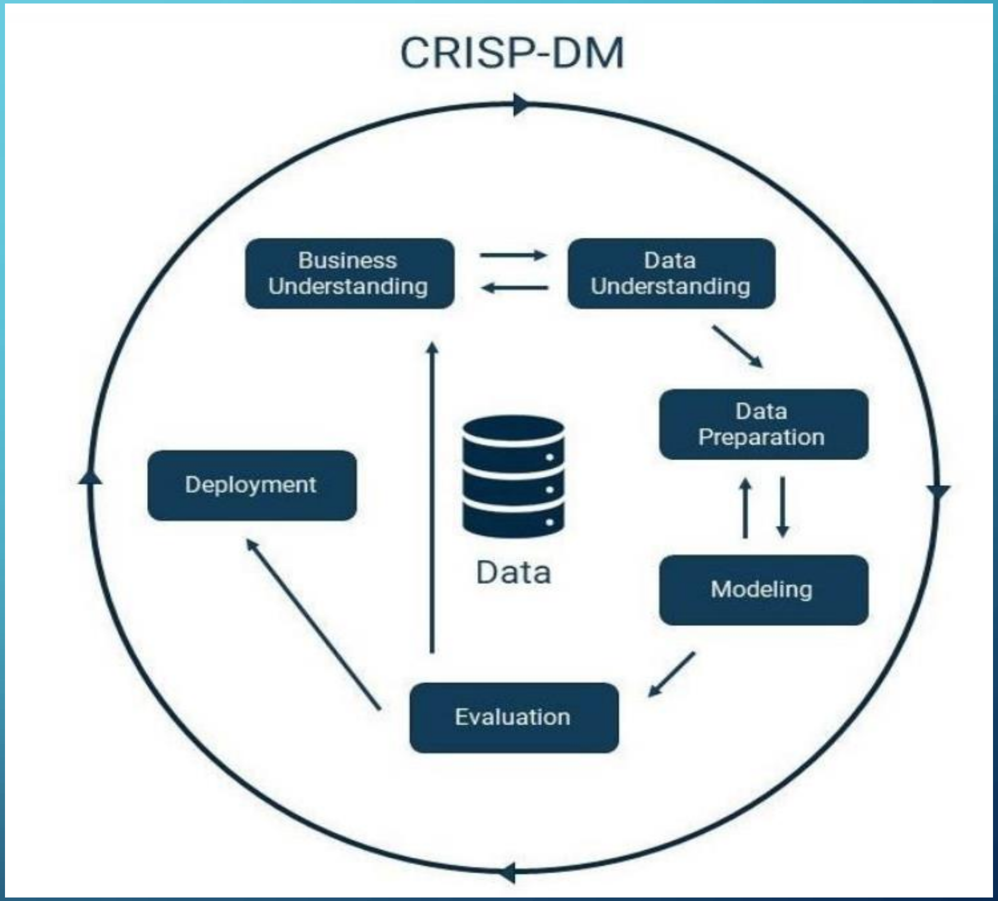
Type 2 diabetes is a chronic condition in which the body becomes resistant to **insulin**, leading to elevated **blood sugar** levels. It's often associated with lifestyle factors such as **obesity**, **poor diet**, and **physical inactivity.**

- **Purpose of this Project:**

The purpose of this project is to develop a predictive model that can accurately identify individuals in Pima Indian tribe who are likely to be diagnosed with diabetes based on various health-related features such as blood glucose levels, body mass index (BMI), number of pregnancies, and other relevant factors.

METHODOLOGY

- This project followed the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to achieve its objectives. This process included the stages of business understanding, data understanding, data preparation, and modelling and used **IBM SPSS Modeler** as its primary tool.



DATA UNDERSTANDING

The dataset contains **768 samples** of women from the Pima Indian Tribe, with the following health-related features recorded:

- **Pregnancies –** Number of times the patient has been pregnant.
- **Glucose –** Plasma glucose concentration measured 2 hours after an oral glucose tolerance test.
- **Blood Pressure –** Diastolic blood pressure (mm Hg).
- **Skin Thickness –** Triceps skinfold thickness (mm).
- **Insulin –** 2-hour serum insulin (mu U/ml).
- **BMI –** Body mass index (weight in kg / height in m^2).
- **Diabetes Pedigree Function –** A function that scores likelihood of diabetes based on family history.
- **Age –** Age in years.
- **Outcome –** Class variable (0 = non-diabetic, 1 = diabetic).

DESCRIPTIVE STATISTICS OF FEATURES

Using the statistics node in IBM SPSS Modeler the descriptive analysis of the features are shown below:

BloodPressure
Statistics

Count	768
Mean	69.053
Sum	53033
Min	0
Max	122
Range	122
Variance	376.637
Standard Deviation	19.407
Standard Error of Mean	0.700
Median	72
Mode	70

Pregnancies
Statistics

Count	768
Mean	3.845
Sum	2953
Min	0
Max	17
Range	17
Variance	11.354
Standard Deviation	3.370
Standard Error of Mean	0.122
Median	3
Mode	1

Glucose
Statistics

Count	768
Mean	120.895
Sum	92847
Min	0
Max	199
Range	199
Variance	1022.248
Standard Deviation	31.973
Standard Error of Mean	1.154
Median	117
Mode	99*

*Multiple modes exist. The smallest value is shown.

BMI
Statistics

Count	768
Mean	31.993
Sum	24570.300
Min	0.000
Max	67.100
Range	67.100
Variance	62.160
Standard Deviation	7.884
Standard Error of Mean	0.284
Median	32.000
Mode	32.000

Insulin
Statistics

Count	768
Mean	79.799
Sum	61286
Min	0
Max	846
Range	846
Variance	13281.180
Standard Deviation	115.244
Standard Error of Mean	4.159
Median	30.500
Mode	0

DiabetesPedigreeFunction
Statistics

Count	768
Mean	0.472
Sum	362.401
Min	0.078
Max	2.420
Range	2.342
Variance	0.110
Standard Deviation	0.331
Standard Error of Mean	0.012
Median	0.372
Mode	0.254*

*Multiple modes exist. The smallest value is shown.

SkinThickness
Statistics

Count	768
Mean	20.536
Sum	15772
Min	0
Max	99
Range	99
Variance	254.473
Standard Deviation	15.952
Standard Error of Mean	0.576
Median	23
Mode	0

Age
Statistics

Count	768
Mean	33.241
Sum	25529
Min	21
Max	81
Range	60
Variance	138.303
Standard Deviation	11.760
Standard Error of Mean	0.424
Median	29
Mode	22

Outcome
Statistics

Count	768
Mean	0.349
Sum	268
Min	0
Max	1
Range	1
Variance	0.227
Standard Deviation	0.477
Standard Error of Mean	0.017
Median	0
Mode	0

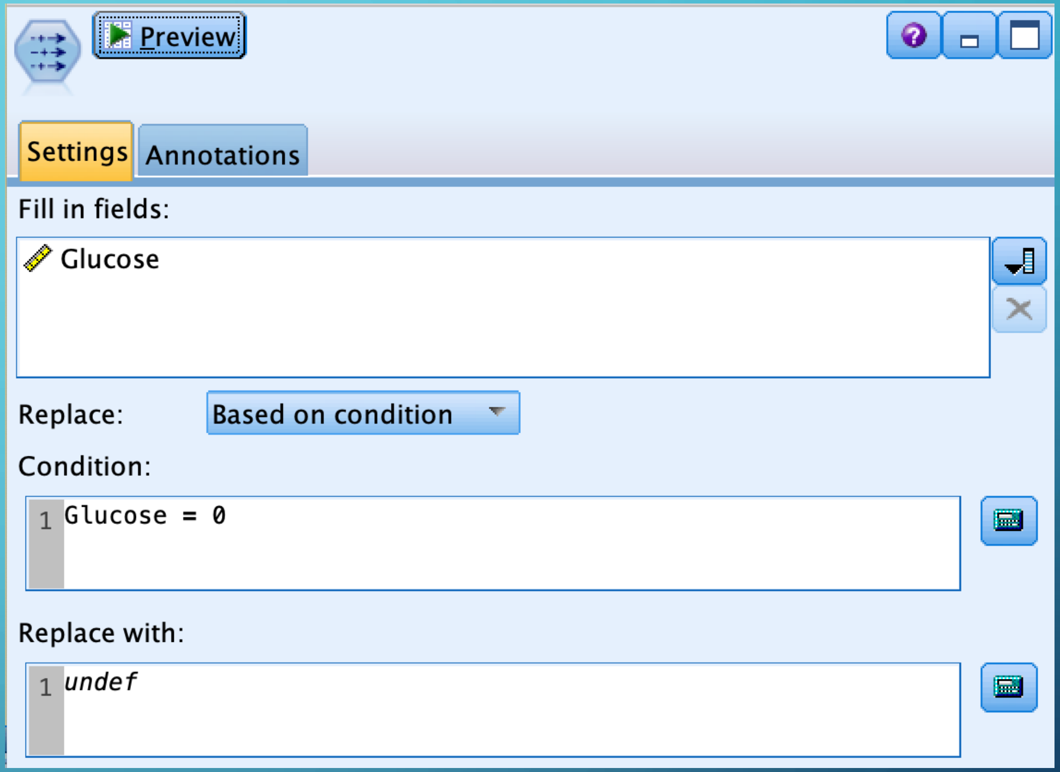
DATA COLLECTION

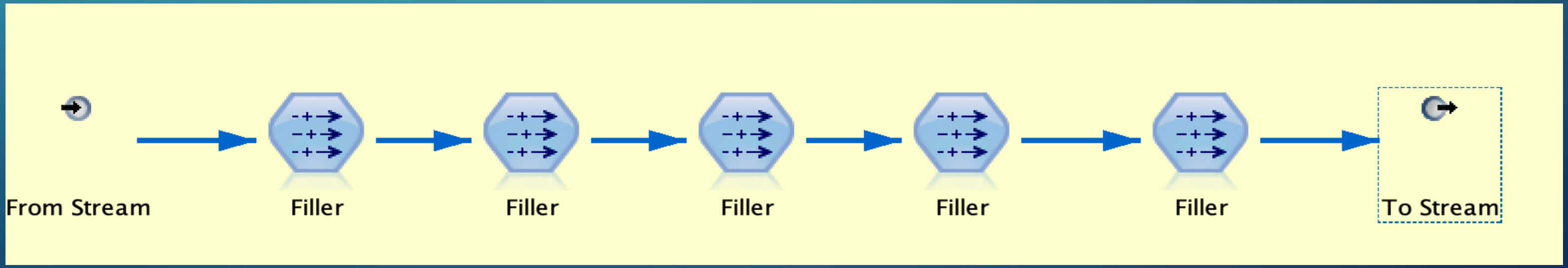
- Using the Var. File node the dataset was added to the modeler as a csv file.
- A type node was then connected to check the data types and their role.
- Outcome was then changed to be a flag features and its role changed from input to target.
- The rest of the features remained as continuous.

Field	Measurement	Values	Missing	Check	Role
Pregnancies	Continuous	[0,17]		None	Input
Glucose	Continuous	[0,199]		None	Input
BloodPress...	Continuous	[0,122]		None	Input
SkinThickn...	Continuous	[0,99]		None	Input
Insulin	Continuous	[0,846]		None	Input
BMI	Continuous	[0.0,67.1]		None	Input
DiabetesPe...	Continuous	[0.078,2....		None	Input
Age	Continuous	[21,81]		None	Input
Outcome	Flag	1/0		None	Target

HANDLING NOISE IN THE DATASET

- For the five features — **Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI**— missing values were initially recorded as **zeros.** Using the **Filler node** in SPSS Modeler, these zero values were **replaced with null values,** which will be imputed with appropriate values.
- For clarity of the workflow, a **Super Node** was created to incorporate all five Filler nodes.





TABLE

A table is created using the table node to view the changes being made to the dataset:

Table | Annotations

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
1	6	148	72	35	$null$	33....	0.627	50	1
2	1	85	66	29	$null$	26....	0.351	31	0
3	8	183	64	$null$	$null$	23....	0.672	32	1
4	1	89	66	23	94	28....	0.167	21	0
5	0	137	40	35	168	43....	2.288	33	1
6	5	116	74	$null$	$null$	25....	0.201	30	0
7	3	78	50	32	88	31....	0.248	26	1
8	10	115	$null$	$null$	$null$	35....	0.134	29	0
9	2	197	70	45	543	30....	0.158	53	1
10	8	125	96	$null$	$null$	$n...	0.232	54	1
11	4	110	92	$null$	$null$	37....	0.191	30	0
12	10	168	74	$null$	$null$	38....	0.537	34	1
13	10	139	80	$null$	$null$	27....	1.441	57	0
14	1	189	60	23	846	30....	0.398	59	1
15	5	166	72	19	175	25....	0.587	51	1
16	7	100	$null$	$null$	$null$	30....	0.484	32	1
17	0	118	84	47	230	45....	0.551	31	1
18	7	107	74	$null$	$null$	29....	0.254	31	1
19	1	103	30	38	83	43....	0.183	33	0
20	1	115	30	30	96	34....	0.529	32	1
21	3	126	88	41	235	39....	0.704	27	0
22	8	99	84	$null$	$null$	35....	0.388	50	0
23	7	196	90	$null$	$null$	39....	0.451	41	1
24	9	119	80	35	$null$	29....	0.263	29	1

DISTRIBUTION AND CORRELATION IDENTIFICATION

- To identify the **data distribution type** and the **strength of correlation** between fields, the **Sim Fit node** was used.
- Subsequently, a **Sim Gen node** was created to generate simulated data based on these distributions.
- The results are shown below:

Simulated Fields: Use Closest Fit   Fit Details...

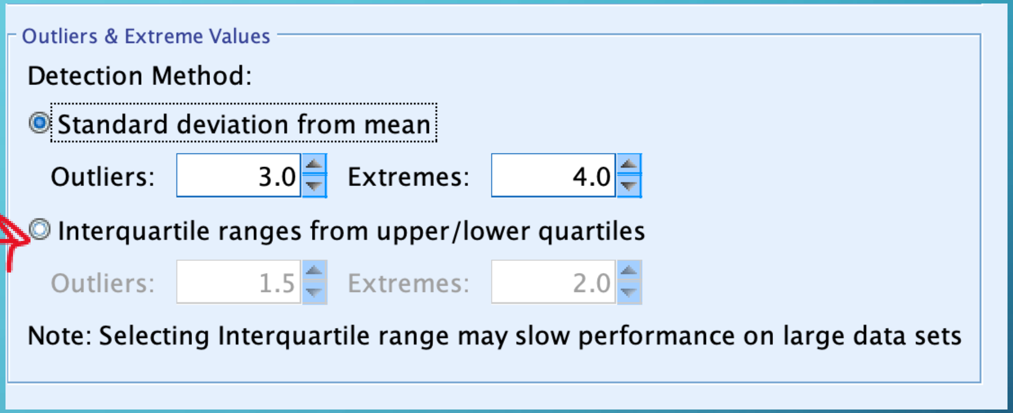
Field	Storage	Status	🔒	Distribution	Parameters	Min,Max
Pregnancies	Integer	✓	☐	Exponential	[scale=0.302936...	[Max=,Min=]
Glucose	Integer	✓	☐	Lognormal	[a=118.8726586...	[Max=,Min=]
BloodPressure	Integer	✓	☐	Normal	[mean=70.5612...	[Max=,Min=]
SkinThickness	Integer	✓	☐	Weibull	[shape1=32.662...	[Max=,Min=]
Insulin	Integer	✓	☐	Lognormal	[a=123.1151029...	[Max=,Min=]
BMI	Real	✓	☐	Gamma	[shape=22.6996...	[Max=,Min=]
DiabetesPedigree...	Real	✓	☐	Lognormal	[a=0.432081207...	[Max=,Min=]
Age	Integer	✓	☐	Lognormal	[a=29.47898864...	[Max=,Min=]
Outcome	Integer	✓	☐	Categorical	[0=0.668367346...	

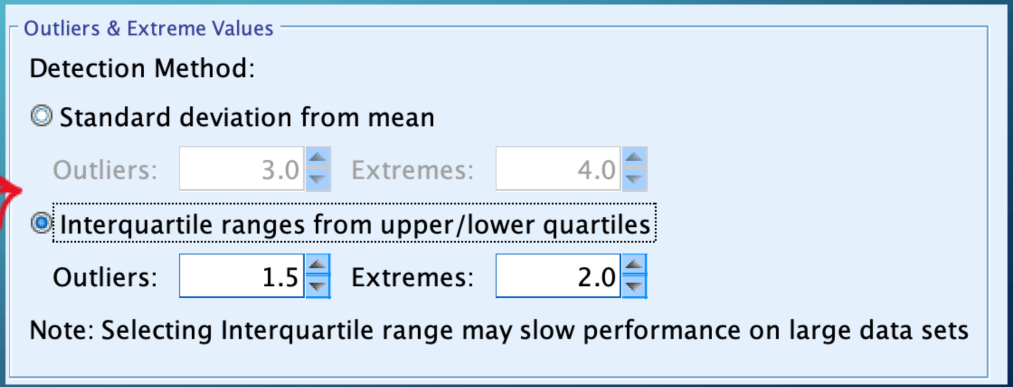
	Age	BMI	BloodPressu...	DiabetesPe...	Glucose	Insulin	Pregnancies	SkinThickne...
Age	1.000	0.070	0.295	0.085	0.344	0.217	0.680	0.168
BMI	0.070	1.000	0.299	0.159	0.210	0.226	-0.025	0.664
BloodPressure	0.295	0.299	1.000	-0.016	0.209	0.101	0.216	0.229
DiabetesPed...	0.085	0.159	-0.016	1.000	0.140	0.136	0.008	0.160
Glucose	0.344	0.210	0.209	0.140	1.000	0.581	0.198	0.199
Insulin	0.217	0.226	0.101	0.136	0.581	1.000	0.079	0.182
Pregnancies	0.680	-0.025	0.216	0.008	0.198	0.079	1.000	0.093
SkinThickness	0.168	0.664	0.229	0.160	0.199	0.182	0.093	1.000

MANAGING EXTREME AND OUTLIER DATA

Detection of Extremes and Outliers:

- Extreme and outlier values were identified for each field based on their **data distribution.**
- Using the **Data Audit node,** detection methods and value ranges were defined.
- **Blood Pressure** followed a **normal distribution** → detected using the **Z-score method:**
  - Outliers: $\pm 3\sigma$ | Extremes: $\pm 4\sigma$
- Other fields were **non-normal** → detected using the **IQR method:**
  - Outliers: 1.5×IQR | Extremes: 2×IQR

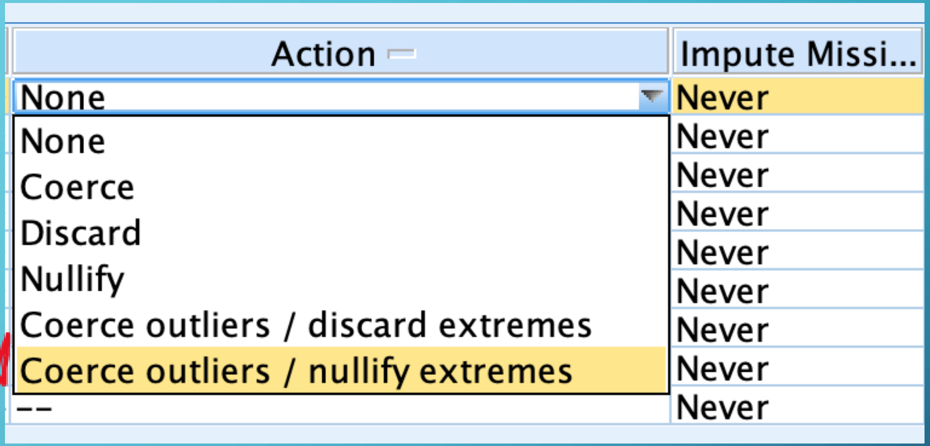




MANAGING EXTREME AND OUTLIER DATA

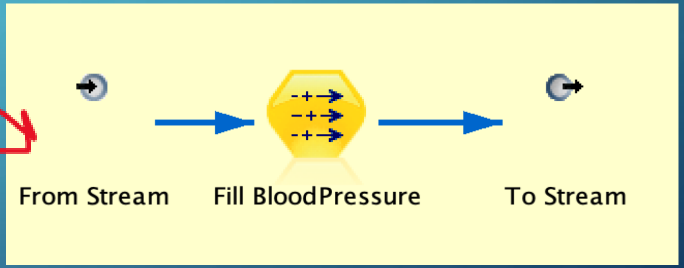
Treatment of Extremes and Outliers

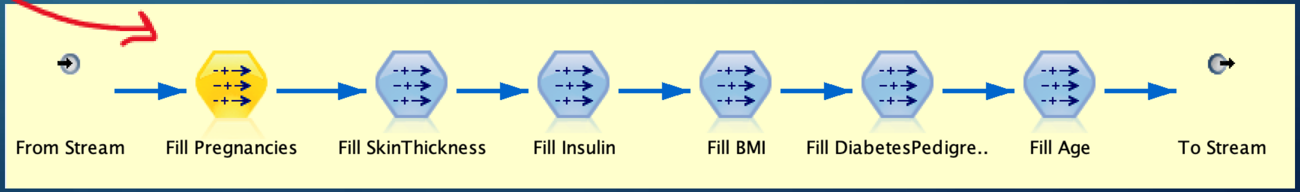
- After detection, the chosen handling method was **"Coerce Outliers and Nullify Extremes."**
- **Outliers** → replaced with the **maximum or minimum** value of their field.
- **Extreme values** → replaced with **nulls**, to be **imputed later** during the missing value handling process.



Super Node Creation

- **Super Node of Filler nodes** were created to incorporate the outlier and extreme value handling processes.
- **Blood Pressure Super Node** – applied the **Z-score method** (normal distribution).
- **Other Fields Super Node** – applied the **IQR method** (non-normal distributions).





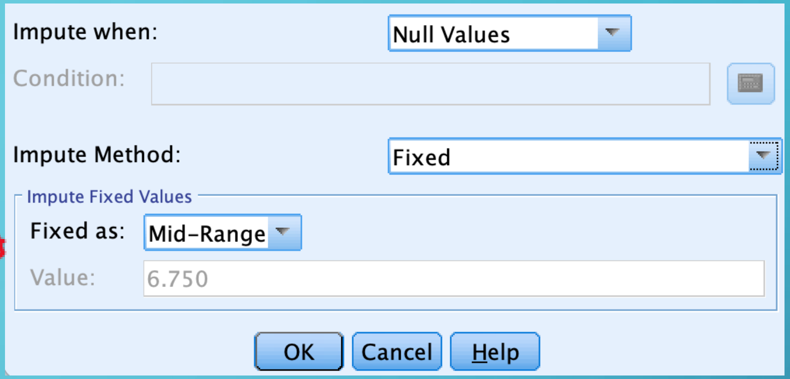
MANAGING MISSING VALUES

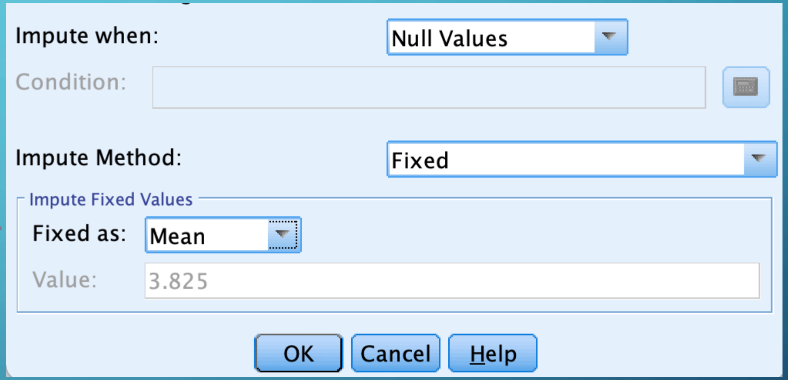
The **Data Audit** node was used to identify and impute missing values appropriately. A **Super Node** was created to integrate these pre-processing steps.

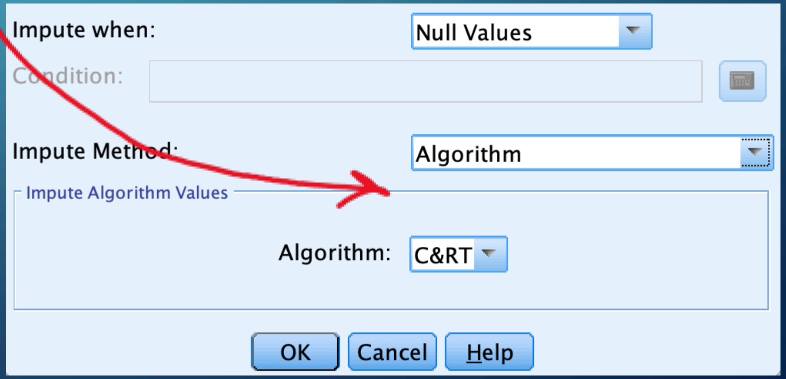
- Missing values in **Pregnancies, Glucose, Diabetes Pedigree Function (DPF)**, and **Age** were imputed using the **median**, as these fields contained skewed distributions.

- Missing values in **Blood Pressure** and **BMI** were imputed using the **mean**, since their distributions were approximately normal.

- For **Skin Thickness** and **Insulin**, the **CART (Classification and Regression Tree)** algorithm was used for imputation, as these fields had a relatively high number of missing values, making simple statistical imputation less reliable.



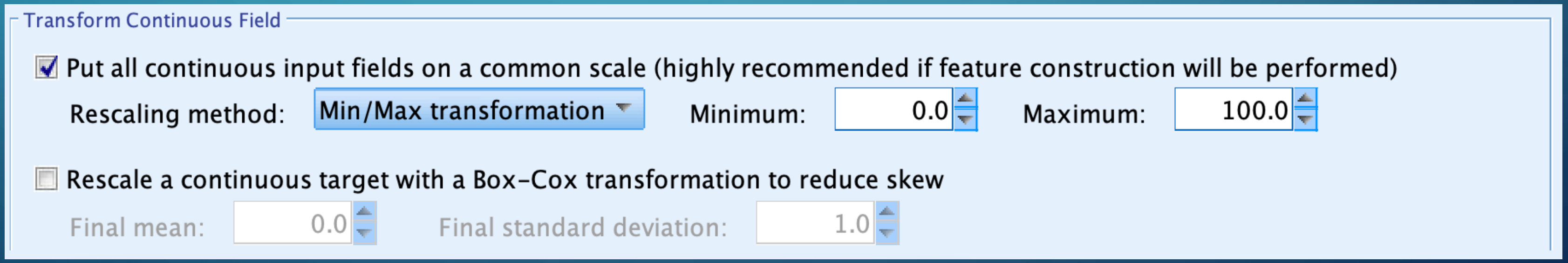




DATA NORMALIZATION

After connecting a **Type** node to the stream (due to changes in data ranges), the **Auto Data Preparation (Auto Data Prep)** node was used to normalize the fields.

- In the *Settings* tab under *Prepare Inputs and Targets*, the **Min–Max transformation** method was selected, with a minimum value of **0** and a maximum of **100.**
- This process scaled all input fields (except the target variable) to fall within the range of 0 to 100.
- Normalization is an essential pre-processing step, as it ensures that all features contribute equally during the modelling phase and prevents features with larger numerical ranges from dominating the model.



MODELLING

- The **K-Means clustering** algorithm was selected as the modelling technique for this project to identify patterns and group individuals based on their health attributes.
- To determine the optimal number of clusters, the **Auto Cluster** node was used, with the **silhouette coefficient** chosen as the evaluation metric for ranking models.
- The Auto Cluster process evaluated models with **1, 2, 3, 4, and 5 clusters**, and the **K-Means model with 2 clusters** achieved the highest silhouette score.

Simple | Expert

Parameter	Options
Number of clusters	1; 2; 3; 4; 5
Generate distance field	false
Show cluster proximity	false
Optimize	Memory

Estimated number of models to be executed: 5

Fields | Model | Expert | Discard | Annotations

Model name: Auto Custom

Use partitioned data

Rank models by: Silhouette

Rank models using: Training partition Test partition

Number of models to keep: 5

Estimated number of models to be executed: 5

Fields | Model | Expert | Discard | Annotations

Models used:

Use?	Model type	Model parameters	No of models
☐	Kohonen	Default	1
☑	K-means	Specify...	5
☐	TwoStep	Default	1

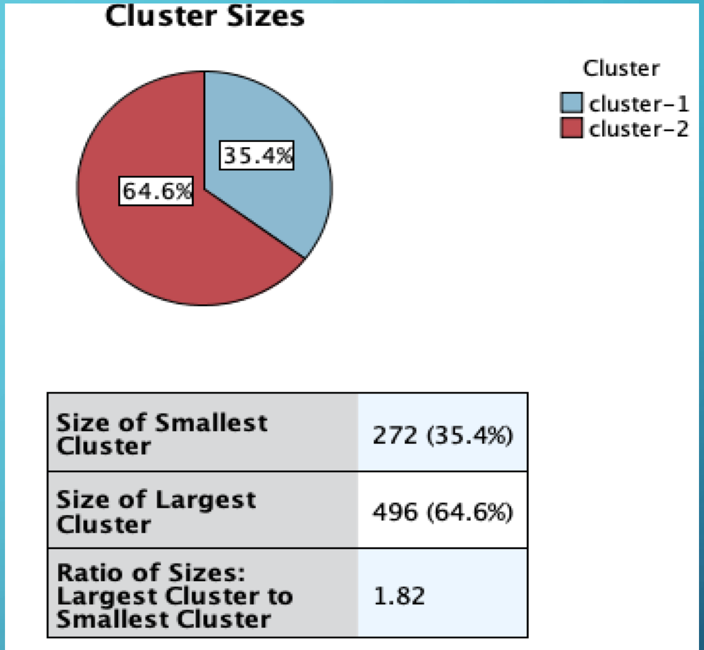
Model | Summary | Annotations

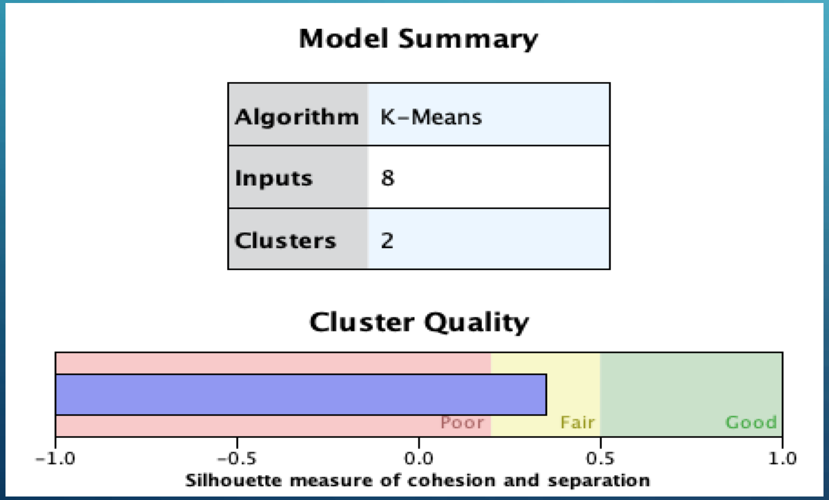
Sort by: Use | Ascending | Descending | Delete Unused Models | View: Training set

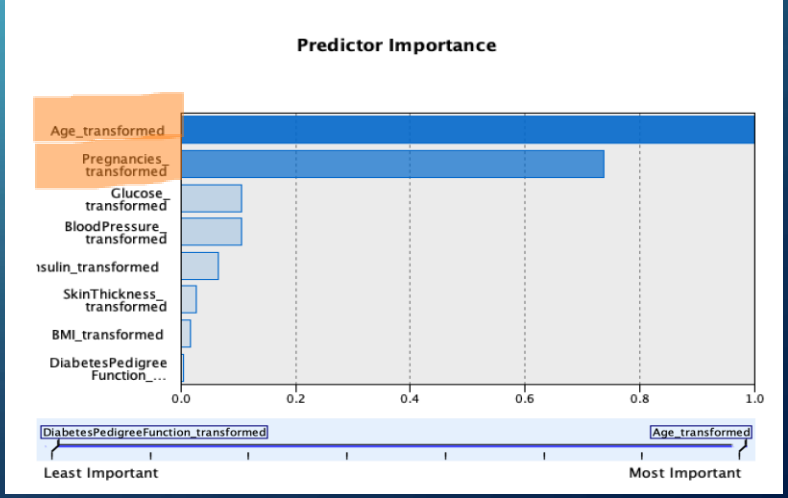
Use?	Graph	Model	Build Time (mins)	Silhouette	Number of Clusters	Smallest Cluster (N)	Smallest Cluster (%)	Largest Cluster (N)	Largest Cluster (%)	Smallest/ Largest	Importan...
☐		K-means 1	< 1	∞	1	768	100	768	100	1.0	0.0
☑		K-means 2	< 1	0.351	2	272	35	496	64	0.548	0.091
☐		K-means 3	< 1	0.318	3	202	26	324	42	0.623	0.172
☐		K-means 5	< 1	0.297	5	93	12	241	31	0.386	0.147
☐		K-means 4	< 1	0.293	4	109	14	279	36	0.391	0.161

MODELLING

- After determining the optimal number of clusters as **2,** a separate **K-Means** node was connected to the stream to generate predictions.
- The clustering process produced **two groups** with proportionate sizes of **35.4%** and **64.6%**, respectively.
- Feature importance analysis revealed that **Age** and **Number of Pregnancies** were the most influential factors in cluster formation, while the **Diabetes Pedigree Function (DPF)** had the lowest importance.
- These results suggest that demographic and reproductive factors play a significant role in differentiating individuals in this dataset, potentially reflecting different levels of diabetes risk.







RESULTS

The table below presents the **mean values** of key parameters for each cluster, along with their **importance levels** as determined by the K-Means model. These statistics help to interpret how each feature contributed to the formation and differentiation of the two clusters.

Cluster 1	Cluster 2
Non-diabetic 64.4 %	Diabetic 35.4 %
Age (importance level=1) Mean=11.8	Age (importance level=1) Mean=54.0
Number of Pregnancies (importance level=0.74) Mean= 15.2	Number of Pregnancies (importance level=0.74) Mean=52.3
Blood Pressure (importance level=0.11) Mean=46.3	Blood Pressure (importance level=0.11) Mean=56.7
Insulin Level (importance level=0.07) Mean=32.6	Insulin Level (importance level=0.07) Mean=41.9

CONCLUSION

- The project used the **CRISP-DM framework** to analyse the *Pima Indians Diabetes Dataset*.
- Data was cleaned, missing values were imputed, and features were normalized before modelling.
- The **K-Means clustering** algorithm identified **2 distinct clusters** in the dataset.
- **Age** and **Number of Pregnancies** were the most influential features; **DPF** had the lowest importance.
- The two clusters likely represent **diabetic** and **non-diabetic** individuals..
- The project highlights how **unsupervised learning** can uncover hidden patterns in health data.
- **Evaluation and deployment** stages were not included, but results show potential for future predictive modelling.